Hard AbsTopK input gate on MNIST: hard backward against DUST steps at five population sizes

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1 Setup

Two-layer MLP on MNIST, $z = W_2\,\mathrm{ReLU}(W_1 x' + b_1) + b_2$ with hidden width 1024, where $x' = \mathrm{AbsTopK}_K(e^g \odot x)$ keeps the K entries of largest magnitude of the rescaled input; $g \in \mathbb{R}^{784}$ is trainable from 0, pixels are in $[0, 1]$. AdamW, learning rate 10^{-3} with 100 warmup steps and a cosine decay to 10^{-4} over 5000 steps of batch 128, no weight decay, no gradient clipping, seed 1337 (the same data order and initialization for every run of a given K).

Hard: the backward passes the hard mask, so g_n moves only through samples whose support contains pixel n. **DUST**: the same gate, with every second optimizer step a DUST step (schedule $[1, 1]$): the gate input $u = e^g \odot x$ of every sample is perturbed on all 784 pixels with $\mathcal{N}(0, \sigma^2)$ noise, $\sigma = 0.1$, in P draws (the population), the tail is rerun, and

$$\hat{g}_t = \frac{1}{P\,\sigma\,B} \sum_{i=1}^{P} \big(\ell_t^{(i)} - \bar{\ell}_t\big)\, a_t^{(i)}, \qquad \hat{G}_g = \sum_t \hat{g}_t \odot e^g \odot x_t$$

is applied to g by the shared optimizer; nothing else moves on a DUST step, so a DUST run has 2500 hard steps for W_1, W_2 against 5000 for the hard run. $K \in \{4, 8, 16, 32\}$, $P \in \{512, 2000, 8000, 32000, 60000\}$. Training CE is the clean hard forward on both kinds of step; test CE and accuracy are the deterministic hard forward on the 10 000 test images.

2 Results

RESULTS-PLACEHOLDER

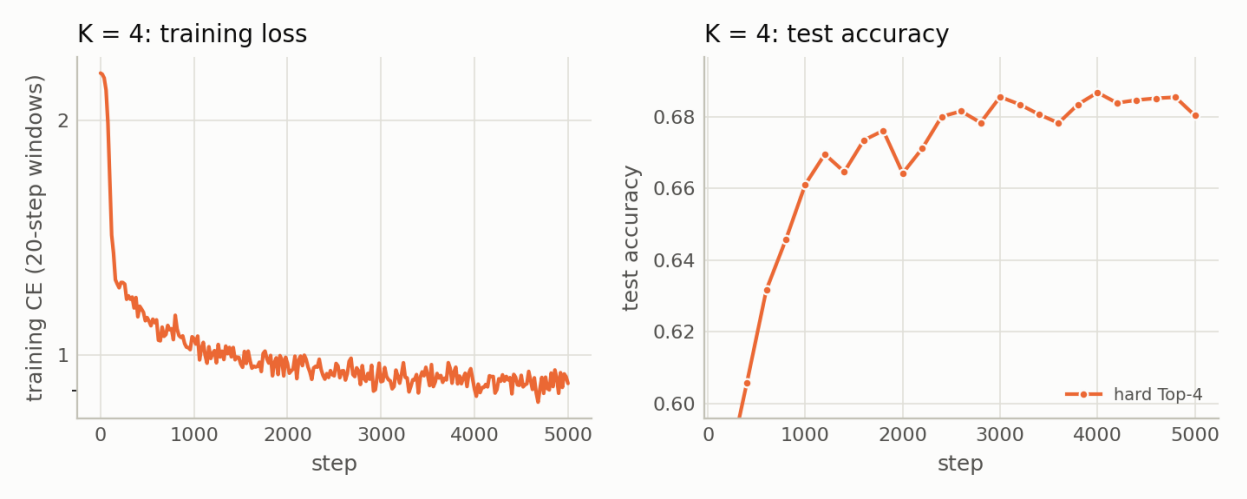


Figure 1: $K = 4$: training CE (left, log scale, 20-step window means) and test accuracy (right, every 200 steps) for the hard backward and DUST at five populations.

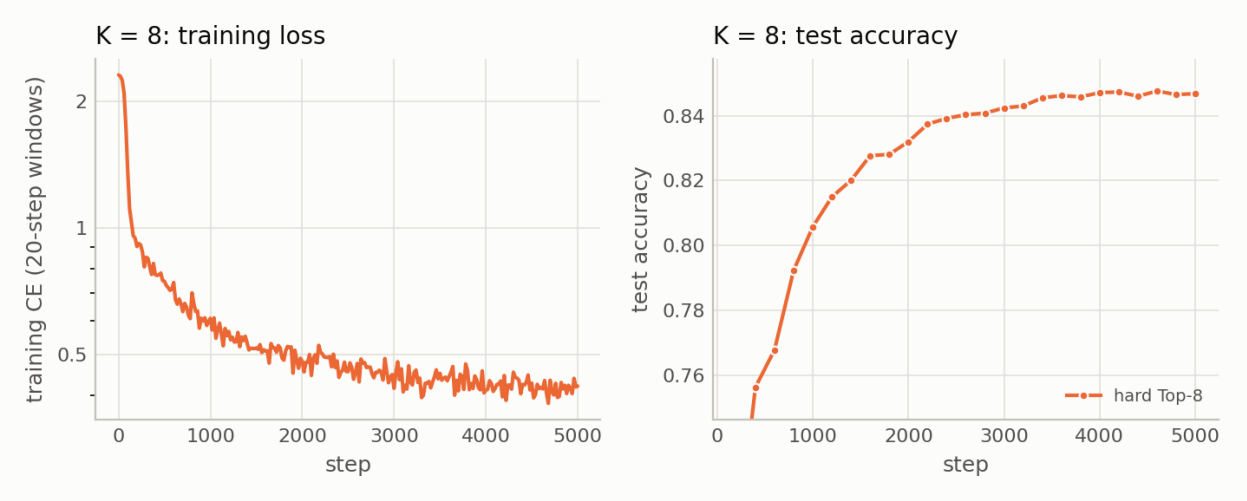


Figure 2: $K = 8$: training CE (left, log scale, 20-step window means) and test accuracy (right, every 200 steps) for the hard backward and DUST at five populations.

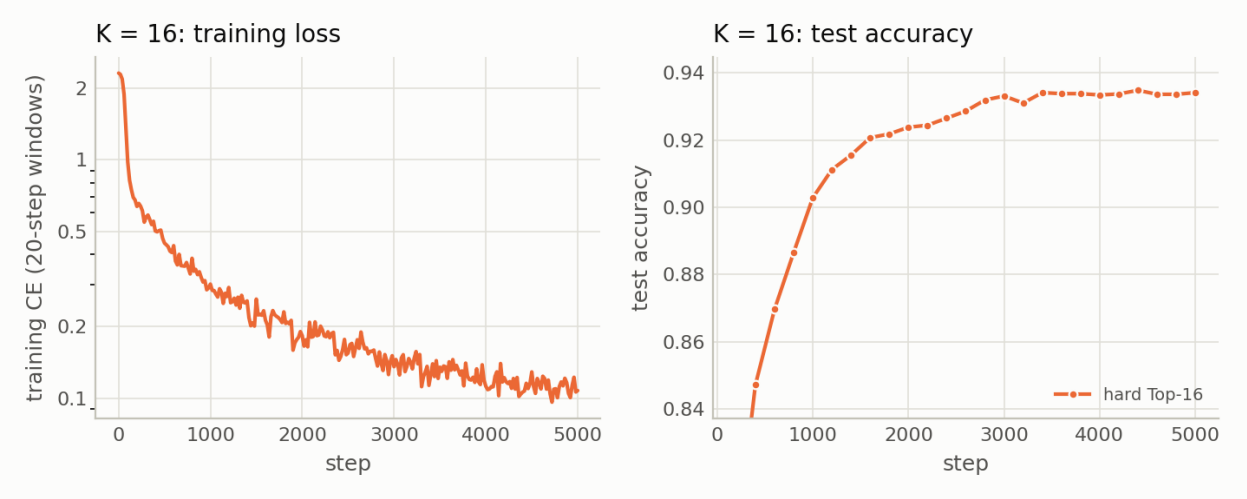


Figure 3: $K = 16$: training CE (left, log scale, 20-step window means) and test accuracy (right, every 200 steps) for the hard backward and DUST at five populations.

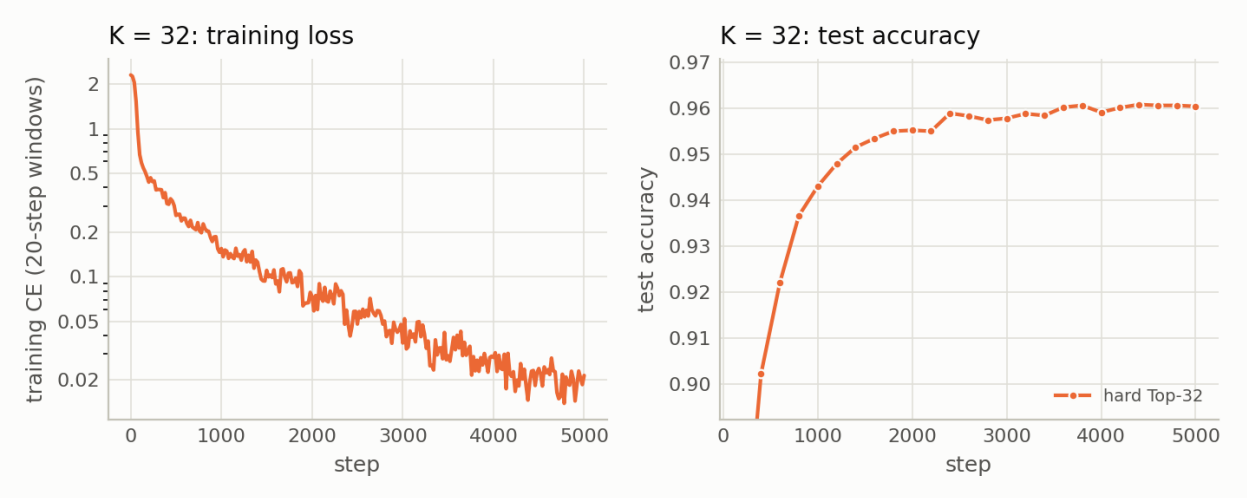


Figure 4: $K = 32$: training CE (left, log scale, 20-step window means) and test accuracy (right, every 200 steps) for the hard backward and DUST at five populations.

Table 1: $K = 4$: values at the final step (5000). Training CE is the mean of the last 20-step window of the clean hard forward; test CE and accuracy are the deterministic hard forward on the 10 000 test images.

method	population	train CE	test CE	test acc (%)
hard Top-K	–	0.9186	0.9305	68.03

Table 2: $K = 8$: values at the final step (5000). Training CE is the mean of the last 20-step window of the clean hard forward; test CE and accuracy are the deterministic hard forward on the 10 000 test images.

method	population	train CE	test CE	test acc (%)
hard Top-K	–	0.4203	0.4834	84.68

Table 3: $K = 16$: values at the final step (5000). Training CE is the mean of the last 20-step window of the clean hard forward; test CE and accuracy are the deterministic hard forward on the 10 000 test images.

method	population	train CE	test CE	test acc (%)
hard Top-K	–	0.1077	0.2174	93.41

Table 4: $K = 32$: values at the final step (5000). Training CE is the mean of the last 20-step window of the clean hard forward; test CE and accuracy are the deterministic hard forward on the 10 000 test images.

method	population	train CE	test CE	test acc (%)
hard Top-K	–	0.0215	0.1466	96.05